

A Guide to Writing Physical Law Triplets

in the Valid Dictionary Framework

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Prepared for the VD Newton project

1. Introduction: Why This Guide Exists

Writing a physical law in the Valid Dictionary framework is not a matter of transcription. You are not taking a known equation and reformatting it. You are making a series of structural decisions — about headwords, anchoring, residual content, peripheral definitions, and ordering — that collectively determine the shape of your theory.

These decisions already exist. Any theoretician encoding a law is navigating them whether they know it or not. The problem is that without a clear map of the decision space, the choices get made unconsciously, inconsistently, and often differently from one entry to the next. The theoretician thinks they are “just writing down the physics,” when in fact they are committing to one of many structurally distinct encodings — and then switching to a different one three entries later without realising it.

Consider anchoring alone. A single law-triplet has three candidate headwords, any one of which might serve as the anchor. With six laws in a theory, that gives $3^6 = 729$ possible anchoring configurations. Each one produces a different dependency graph, different peripheral entries, and different attachment points for later laws. These 729 theories cover the same physics, but they are not the same object. They have different structures, different pedagogical orderings, and different strengths and weaknesses.

If you try to work with an abstract version of the theory — one where you imagine all anchoring choices are equivalent and try to engineer all of them simultaneously — you will run in circles. You will write a wall that makes sense under one anchoring, then try to use it as a chimney for a triplet that assumes a different one. The entries will refuse to fit together, and the source of the friction will be invisible, because you never identified the choice you were making.

The purpose of this guide is to make the decision space explicit. Each section identifies a specific choice the theoretician must make, explains what constrains it, and describes how it interacts with the other choices. The guide does not tell you which decisions to make — those depend on your subject matter, your audience, and your goals. It tells you that the decisions exist, so that you can make them deliberately rather than by accident.

2. Estimating the Law Structure of a Theory

The problem

Before you begin encoding, you need an estimate of what you are looking for. How many triplets does this theory contain? Which concepts are locked together by laws, and which are derived consequences? Where should you expect the irreducible mutual dependencies to live?

The natural place to start is the historical record — textbooks, foundational papers, the received list of “laws” and “principles” that define the field. But the historical record is an unreliable guide to triplet structure. The word “law” has meant different things at different times, genuine laws hide behind other labels, and the textbook’s count of laws is rarely the count of triplets. This section helps you navigate these difficulties and arrive at a working estimate.

What “law” has historically meant

The word “law” has been applied to at least four different kinds of scientific statement, and only one of them corresponds to a VD triplet.

Empirical regularities. Kepler’s laws are the prototype. They describe patterns extracted from observational data — the planets move in ellipses, the periods relate to the semi-major axes. These are relationships among quantities that were already defined. They introduce no new concepts and lock nothing together. In VD terms they are constraints or theorems, not triplets. A statement that can be checked against data using only pre-existing vocabulary is almost certainly not a law in the VD sense.

Philosophical necessities. Descartes’s laws of motion are derived from metaphysical first principles — the immutability of God, the conservation of the “quantity of motion.” Here “law” means something deduced from a philosophical commitment. These may or may not contain triplet structure, but the metaphysical framing makes it difficult to isolate. The concepts are entangled with a worldview rather than with each other in the precise, auditable way the VD requires.

Structural commitments. Newton’s laws are the turning point. They introduce new technical concepts — force, inertial mass, inertial frame — and define them through their mutual relationships. The concepts did not exist before the law and cannot be understood independently of each other. This is what a VD triplet looks like from the outside. When “law” is used in this sense — as a statement that simultaneously introduces and interlocks new concepts — it is a reliable indicator of triplet structure.

Derived consequences. Many statements called “laws” are theorems. Conservation of energy, conservation of momentum, Gauss’s law in electromagnetism — these can be derived from more fundamental commitments. They are important, but they are not triplets. They do not introduce irreducibly new concepts. If a “law” can be proved from other laws, it is downstream, not foundational.

The modern aliasing problem

After Newton, the usage of “law” became inconsistent and eventually fell out of fashion. Different fields adopted different labels for their foundational commitments.

Thermodynamics still says “laws” — the zeroth, first, second, and third laws. Electromagnetism says “equations” — Maxwell’s equations. Quantum mechanics says “principles” and “postulates” — the exclusion principle, the uncertainty principle, the superposition principle, the Born rule. Relativity says “postulates” and “principles” — the equivalence principle, the principle of relativity.

These labels do not track a real distinction. A “principle” in quantum mechanics may be doing exactly the same structural work as a “law” in thermodynamics — introducing new concepts and locking them together in an irreducible cycle. The theoretician must look past the label and ask: does this statement introduce genuinely new vocabulary that is mutually defined? If yes, it is a candidate triplet regardless of whether the field calls it a law, a principle, a postulate, or an assumption.

The shift from “law” to softer language reflects a growing discomfort in physics with the philosophical weight of “law” — its connotations of necessity, universality, and legislation. But the structural phenomenon the VD captures did not change. Laws kept being produced; they just stopped being called laws.

The bundling problem

Even when the historical record correctly identifies a law, it may package the structure incorrectly.

Newton’s Third Law is the clearest example. Traditionally presented as a single law — “every action has an equal and opposite reaction” — the VD encoding revealed it to be two separate degree-4 laws, each with its own triplet, its own anchor, and its own winters. The textbook count says one law. The triplet count says two.

The reverse also occurs. Statements spread across multiple textbook “laws” or “axioms” may turn out to encode a single triplet — the three statements are the three corners of one law, artificially separated in the pedagogy.

The textbook’s count of laws is therefore neither an upper bound nor a lower bound on the number of triplets. It is a rough hint, subject to both splitting and merging.

The winter-word heuristic

A practical way to estimate the number of triplets is to count the genuinely new technical terms the theory introduces — the terms that cannot be defined without circular reference to each other.

Every triplet introduces exactly two winter words. So if a theory introduces roughly $2N$ irreducibly new concepts, you are probably looking at roughly N triplets. This gives a sanity check: if the textbook lists five “laws” but the theory only introduces four new concepts that resist independent definition, something is off — some of

those “laws” are probably theorems, or some of the textbook’s “laws” are corners of the same triplet.

This heuristic is rough. It requires you to already have a sense of which terms are genuinely new and which are derived, which is part of what the encoding is supposed to discover. But it gives a starting range to work within, which is better than trusting the textbook count blindly.

The resolution degrades with depth

The historical record is most helpful for low-degree laws. Degree-1 and degree-2 laws tend to involve concepts that are intuitive, well-studied, and clearly named.

Physicists and philosophers have had centuries to argue about force, mass, acceleration, temperature, entropy. The traditional analysis of what counts as a “law” was built with these cases in mind, and it works reasonably well for them.

At higher degrees, the record becomes sparse. Degree-4 laws and above involve concepts that depend on several layers of prior law-structure before they can even be stated. These concepts are more abstract, more technical, and less likely to have been singled out and named in the historical literature. The traditional analysis of lawhood tends to stop at about this depth — not because the structure ends, but because the philosophical tools were not designed to track it.

In practice, this means the theoretician should expect diminishing returns from the historical record as they work deeper into the dependency graph. The textbook and the philosophical literature will give good leads for the first few laws. After that, the VD encoding itself becomes the primary tool of discovery. The deeper structure is found by following the dependency graph, not by consulting the historical list.

This is not a failure of the historical record. It is a natural consequence of the fact that the concept of “law” was never formalised in a way that could track depth. The VD provides that formalisation. But it means the theoretician must be prepared to find laws that nobody has called laws before — structural commitments hiding in definitions, derivations, or unstated assumptions, visible only when the dependency graph is drawn.

3. A Note on the Scaffold

What this guide assumes

This guide covers the writing of physical law triplets — the six-entry units that encode laws. It assumes that the kinematic scaffold beneath the laws already exists: the non-law entries that define the mathematical and observational vocabulary the laws will draw on. Position, velocity, acceleration, interval, frame — whatever your theory needs as raw material before the first law is laid down.

Building a good scaffold is its own problem, and it is not trivial. But it is a different problem from law-writing, and this guide does not address it.

How the scaffold shapes the anchor choice

What the guide does need to acknowledge is that the scaffold and the anchor choice are coupled. The anchor term is whichever headword can receive a peripheral definition — a chimney — using terms already in the log. “Terms already in the log” means, in large part, the scaffold.

A richer scaffold gives you more options. If your scaffold defines many kinematic and observational concepts, more candidate headwords will be connectable to prior entries, and you will have genuine freedom in choosing the anchor. A minimal scaffold constrains you — perhaps only one of the three candidate headwords has a natural chimney, and the anchor is effectively forced.

This means that scaffold design and anchoring strategy are not independent. If you find yourself unable to anchor a law the way you want to, the problem may not be with the law — it may be that the scaffold does not contain the concept you need as a bridge. The fix is to go back and add scaffold entries, not to force an unnatural anchor.

Equally, if you know in advance which anchoring strategy you want for your laws, you can design the scaffold to support it. The scaffold is not neutral infrastructure. It is the terrain on which the anchor choices are made, and it can be shaped to make good choices available.

4. Definition of a Physical Law

Formal statement

A physical law in the Valid Dictionary is a set of three entries in the log:

$$E_i \quad A := \text{Res}(B, C)$$
$$E_j \quad B := \text{Res}(A, C)$$
$$E_k \quad C := \text{Res}(A, B)$$

where $i < j < k$, but the indices need not be consecutive — other entries may appear between them in the log.

A, B, and C are headwords. $\text{Res}(X, Y)$ denotes a residual definition parameterised by headwords X and Y: a sentence whose purpose is to produce a specific causal effect in the interpreter, and whose content is expressed in terms of the two named headwords rather than resolving into prior dictionary entries.

The temporal constraint

The three headwords are not symmetric. Exactly one of A, B, C must be previously defined at the stage the first triplet entry is laid down. The other two must be previously undefined at that stage.

The previously defined headword is the November word (also called the anchor). It has at least one earlier entry in the log — a conventional, non-residual definition that resolves into prior headwords. This earlier entry is the chimney.

The two previously undefined headwords are the winter words (also called the cyclic terms). They are born in the triplet. Their first appearance anywhere in the log is as headwords of triplet entries.

Any minimal 3-cycle in the dependency graph that satisfies this temporal constraint is a law. Any minimal 3-cycle that does not satisfy it is not a law. There is no additional recognition step.

What the entries say

Each triplet entry defines its headword as a residual function of the other two. This means:

The definition of A says what A is in terms of how B and C relate through it. The definition of B says what B is in terms of how A and C relate through it. The definition of C says what C is in terms of how A and B relate through it.

No single entry states how any two of the headwords relate to each other directly. That relationship is emergent — it arises from the intersection of all three residual definitions taken together.

Because these definitions are residual, they do not resolve into earlier dictionary content. They bottom out in the causal effect of their wording on a competent interpreter. This is what makes the triplet a law rather than a chain of conventional definitions: the mutual dependence is irreducible. You cannot define any one of the three headwords without invoking the other two.

Roles are identified, not assigned

The November word is not chosen by the dictionary author. It is identified by the temporal constraint: whichever of the three headwords was previously defined is the November word. The two winter words are whichever were not. The roles fall out of the structure of the log automatically.

Non-consecutiveness

The three entries E_i , E_j , E_k must appear in order ($i < j < k$), but the log may contain other entries between them. This is important in practice because the dictionary often needs auxiliary definitions — scaffolding entries, peripheral definitions for other purposes, or entries belonging to other overlapping six-entry units — that fall between the members of a single triplet.

The triplet is identified by the 3-cycle in the dependency graph, not by adjacency in the log.

The complete functional unit

The triplet alone is not self-sufficient. To be fully operational, each law requires six entries: the chimney (a peripheral definition of the November word, appearing before the triplet), the three triplet entries, and two walls (peripheral definitions of the two winter words, appearing after the triplet). The walls give the winter words conventional, outward-facing definitions so that downstream entries can reference them without entering the law’s internal cycle.

The full structure is therefore:

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E_a    A := Def(...)    ← chimney (peripheral, November word)
...
E_i    A := Res(B, C)   ← triplet
E_j    B := Res(A, C)   ← triplet
E_k    C := Res(A, B)   ← triplet
...
E_p    B := Def(...)    ← wall (peripheral, winter word)
E_q    C := Def(...)    ← wall (peripheral, winter word)

```

where $\text{Def}(\dots)$ denotes a conventional definition that resolves into previously defined headwords, and $a < i < j < k < p, q$.

5. Inward-Facing and Outward-Facing Entries

Two kinds of definitional work

Each headword in a law receives two definitions at different points in the log. These definitions face in opposite directions.

Outward-facing entries are the peripheral definitions — the chimney and the walls. They connect the headword to the rest of the dictionary. Their job is to say

what the term is in relation to concepts that already exist (chimney) or to make the term available for future use (walls). They resolve into prior headwords. They are conventional dictionary entries and can be read and understood without knowing anything about the law they belong to.

Inward-facing entries are the triplet definitions. They connect the headword to the other two members of the law. Their job is to say how the other two headwords relate through this one. They are residual — they do not resolve into prior dictionary content but instead produce a causal shift in the interpreter's modelling behaviour. They can only be understood in the context of the law they belong to.

Why the distinction matters in practice

The most common mistake when writing a law is to blur these two roles. The theoretician writes a triplet entry that tries to do outward-facing work — connecting the headword to the broader dictionary — when it should be facing inward toward the other two members of the triplet. Or they write a wall that tries to encode the law-relationship, when it should be a standalone definition.

A useful test: take a peripheral entry and read it in isolation. It should be intelligible without knowing that a triplet exists. It should read like a normal dictionary definition. If it only makes sense in the context of the law, it is doing inward work and belongs in the triplet, not in the periphery.

Conversely, take a triplet entry and ask whether it says anything about the broader dictionary. It should not. It should speak only about the other two headwords in the triplet. If it reaches out to reference terms beyond A, B, and C, it is doing outward work and that content probably belongs in a peripheral entry instead.

The division of labour

This gives a clean principle for distributing content across the six entries:

The chimney says what the November word is, using terms already in the log. It tells the reader: here is a concept you can understand with what you already know.

The triplet entries say how the three headwords lock together. They tell the reader: here is a mutual dependence that cannot be broken apart.

The walls say what the winter words are, using terms available after the triplet. They tell the reader: here are two new concepts, now made available for everything that comes next.

If you find yourself wanting to put too much into a triplet entry, move the excess into a wall. If you find a wall that seems to encode part of the law, move that content into the triplet. The six entries should divide the work cleanly between connecting (peripheral) and locking (triplet).

6. What Each Triplet Entry Says

The rule

Each triplet entry defines its headword in terms of the other two. But there is a specific constraint on what “in terms of” means here. The entry at each corner relates the two daughter headwords to the headword being defined. It does not relate the two daughters to each other.

To make this concrete: the entry for A says how B and C each connect to A. It does not say how B connects to C. That relationship is not stated anywhere in the triplet explicitly. It is emergent — it arises from reading all three entries together and noticing how the connections through A, through B, and through C jointly imply a relationship between any pair.

Why this constraint exists

If a single triplet entry tried to state the relationship between the two daughters directly, it would be doing too much. It would be encoding the whole law at one corner, making the other two corners redundant. The irreducibility of the triplet — the fact that you need all three entries to understand the law — depends on each entry being partial. Each corner gives you a view of the law from one vantage point, and no single vantage point is complete.

This is also what makes the 3-cycle minimal. A 2-cycle would give you two entries each fully defining the other, which collapses into synonymy. A 3-cycle works precisely because each entry is incomplete on its own. The incompleteness is the structure.

What this looks like in practice

Suppose your law has headwords A, B, C with A as the November word. Then:

The entry for A should say something like: “A is the [relationship to B] under conditions involving [relationship to C].” It characterises A by how B and C each bear on it.

The entry for B should say something like: “B is the [relationship to A] that [relationship to C].” It characterises B by how A and C each bear on it.

The entry for C should say something like: “C is the [relationship to A] given [relationship to B].” It characterises C by how A and B each bear on it.

None of these entries says “B is [something] of C” or “C [does something] to B” directly. The B-C relationship is what you reconstruct after reading all three.

A common failure mode

The temptation is strong to write the entry for A — the November word, the familiar concept — in a way that states the whole law. The theoretician knows the law, knows what A “really means” in the context of B and C, and wants to say it all at once. The result is an A-entry that relates B to C through A, rather than relating B and C separately to A.

When this happens, the B and C entries become afterthoughts — restatements or trivial consequences of what A already said. The triplet loses its three-cornered structure and collapses into one dominant entry with two appendages. If your B and C entries feel like they are not doing real work, check whether your A entry has absorbed the relationship that should be emergent.

7. Residuals

What a residual is

Every chain of definitions in a Valid Dictionary must eventually bottom out. The entries where this happens are called residuals. A residual is an entry whose definition string does not resolve into further headwords — it terminates in something that a competent interpreter is expected to be able to do, recognise, or undergo without further dictionary assistance.

Residuals are where reality enters the formal system.

Meaning is involuntary

To understand why residuals work the way they do, you need to accept a claim about meaning. In the VD framework, the meaning of a word is the total causal change produced in an interpreter by the physical event of encountering that word. This is a physical phenomenon, not a contract.

Consider the symbol “+”. When a CPU encounters this token, it performs addition. It cannot choose otherwise — the causal chain from symbol to operation is deterministic. A human encountering “+” is in a structurally similar position: the symbol activates pattern recognition, loads the concept of addition, and primes the person to add. The human may choose not to carry out the addition, but the cognitive activation has already occurred. The meaning — the causal effect of encountering the symbol — has already landed.

The difference between the CPU and the human is one of probability, not of kind. The CPU adds with probability 1. The human adds with very high probability. In neither case is the meaning a matter of agreement or interpretation. It is a physical event triggered by a physical stimulus.

This matters for residuals because it tells you what a residual entry is actually doing. When a reader encounters a residual, the wording of that entry produces a causal shift in their modelling behaviour. The residual is successful if that shift is the one the theory requires. It is unsuccessful if the reader walks away with the wrong disposition — even if the wording is formally unobjectionable.

The reader is assumed to be familiar with the reasoning behind this claim, as developed in the companion note *The Factorial vs Sum Example*.

The two roles of a residual

Because meaning is a causal effect, residuals can serve two distinct roles in a dictionary.

Role 1: Empirical residual. The entry stands for whatever causal effect its physical manifestation actually produces in interpreters. This is a fact about the world. If you wrote a particular sentence and showed it to a hundred physicists, the distribution of resulting modelling behaviours is the empirical meaning of that residual. You can study it, measure it, and refine the wording to narrow the distribution.

Role 2: Idealised residual. The entry stands for a target causal effect that the dictionary author wants to produce, which may not yet have a known best wording. The author knows what modelling behaviour the theory needs at this node — they just haven't yet found the sentence that reliably triggers it. The entry is a specification, not a finished implementation.

The placeholder notation: `E_n.Residue()`

When reasoning about the structure of a dictionary, it is often useful to refer to the idealised role of a residual without committing to specific wording. The notation `E_n.Residue()` serves this purpose.

`E_n.Residue()` is a formal handle for the causal effect that entry n is required to produce. It allows you to discuss what properties the residual must have for the theory to work, without getting tangled in prose. It allows you to compose residuals into larger arguments — if entry m depends on entry n , you can write about `E_n.Residue()` feeding into entry m without specifying either sentence. And it allows you to evaluate candidate wordings against the specification: given a proposed sentence for entry n , you can ask whether this string reliably produces `E_n.Residue()` in a competent interpreter.

This is analogous to declaring a function signature before writing the function body. The theory constrains what the residual must do; the actual wording is the implementation. Separating these two concerns prevents a common failure mode

where an author endlessly tweaks prose without a clear specification of what the prose is supposed to achieve.

How to judge a residual's wording

A candidate sentence for a residual is good if it reliably produces the intended causal effect. Concretely, ask:

Does the sentence tell the interpreter what to do, trust, or attend to? Does it load the correct modelling stance — not merely a formally compatible one, but the one the theory actually needs? Would a competent reader, encountering this sentence in context, be disposed to behave in the way the theory requires at this node?

If your residual wording is ambiguous between two readings that lead to different modelling actions, the wording is not yet good enough — even if both readings are technically consistent with the surrounding definitions.

8. Choosing the Anchor Term

Why this choice matters

Of all the decisions a theoretician makes when encoding a law, the choice of anchor term is the most consequential. It determines which concept serves as the bridge between the new law and everything that came before it in the dictionary. It fixes which two concepts are born as winter words. It constrains what the chimney and walls must say. And because walls from one law become available as chimneys for later laws, the anchor choice propagates forward through the entire dictionary.

In the long run, the most compact way to characterise a particular theory — a specific instance of a Valid Dictionary — may simply be to list the anchor terms of its laws, in order. Two dictionaries covering the same physical domain but choosing different anchors will produce structurally different logs, different dependency graphs, and different pedagogical experiences for the reader. The anchor sequence is, in a sense, the fingerprint of a theory.

What constrains the choice

The anchor must be a term that can receive a peripheral definition — a chimney — using only concepts already present in the log. This is a hard constraint. If none of the three candidate headwords can be connected to prior entries, the law cannot be laid down yet. If only one can, the choice is forced. The interesting case is when two or all three of the headwords could plausibly serve as anchors, and the theoretician must decide between them.

Several considerations bear on this decision.

Historical precedent and order of discovery. The history of a scientific field often reveals a natural ordering of concepts. The concepts that were understood first — or at least named and operationalised first — tend to have richer connections to elementary vocabulary. Following the historical order of discovery is not mandatory, but it is a useful heuristic: if generations of physicists found it natural to start from concept A and arrive at concepts B and C, there is probably a genuine dependency structure underneath that history. Departing from it should be a deliberate choice, not an accident.

Natural connectibility to earlier concepts. Some terms have obvious peripheral definitions in terms of what the dictionary already contains. Others require elaborate scaffolding before they can be defined standalone. If one candidate headword can be given a clean, short chimney while the others would need several preparatory entries, that is evidence in favour of making it the anchor. The chimney should feel like a natural extension of the existing log, not a forced construction.

Downstream consequences. The two winter words will need walls — peripheral definitions that connect them outward to the rest of the dictionary. The theoretician should think ahead: will the walls for these winter words be easy to write? Will they provide useful attachment points for later entries? A choice of anchor that produces winter words with natural, productive walls is better than one that leaves the winters stranded — formally defined within the triplet but hard to connect to anything else.

Simplicity of the peripheral entries. The chimney and walls are conventional definitions. They should be clear, concise, and independently intelligible. If choosing a particular anchor forces you to write a chimney that is convoluted or that smuggles in assumptions, that is a warning sign. Similarly, if the resulting walls are awkward or require extensive qualification, a different anchor choice may produce a cleaner overall structure. The goal is not to minimise the difficulty of any single entry, but to distribute the definitional work evenly across the six entries so that none of them is doing more than its share.

The role of the audience

The anchor choice is not purely a matter of logical structure. It is also a decision about communication. The anchor is the reader's point of entry into the law — the concept they already understand, through which the two new concepts will be introduced. Different audiences bring different prior knowledge.

A dictionary written for experimental physicists might anchor a law in an observable quantity, because that is what the audience has direct experience with. The same law, written for mathematicians, might anchor in a structural property, because that is where the mathematician's prior definitions are richest. Neither choice is wrong. They produce different dictionaries — different theories in the VD sense — optimised for different readers.

This means that the anchor choice is one of the places where the theoretician's judgment about their audience enters the formal structure of the theory. It is not decoration or pedagogy layered on top of the real content. It is a structural commitment that shapes the dependency graph itself.

9. When the Law Doesn't Look Like a Triplet

The claim

The framework says that a physical law is a triplet — exactly three headwords in a minimal 3-cycle. This is not a soft guideline. It is a structural claim about what a law is. If your candidate law appears to have two critical terms, or four, or more, the framework does not bend to accommodate it. Instead, it asks you to look again.

Two terms: find the hidden third

If you believe you have a law relating just two concepts, A and B, then what you have is a 2-cycle: A is defined in terms of B, and B is defined in terms of A. This is a synonym pair. It collapses — the two words are doing the same definitional work, and the cycle contains no irreducible structure.

But real physics does not produce synonym pairs. If A and B feel like they are locked together by a law, something else is present that you have not yet named. There is a third concept C hiding inside one of your definitions, doing load-bearing work without its own headword.

The task is to find it. Look at your definition of A in terms of B. Is there a qualifier, a condition, a mode of relationship that is doing real work in that sentence but has not been given a name? That qualifier is your candidate for C. Pull it out, give it a headword, and check whether A, B, C now form a genuine triplet.

This is often difficult. The hidden third term is typically something the theoretician considers obvious or not worth naming — a background condition, a mode of interaction, a frame of reference. The VD insists that if it is doing definitional work, it needs a headword. The discomfort of naming the obvious is a feature, not a bug: it forces the theoretician to commit to exactly what mediates the relationship between A and B.

Four or more terms: decompose

If you believe you have a law relating four concepts A, B, C, D, then the framework's claim is that you do not have one law. You have two (or more) laws whose triplets overlap, entangled together in a way that makes them look like a single four-term structure.

The task is to find the decomposition. An algebraic analogy makes the procedure concrete.

The functional analogy. Think of a law as a function. A triplet is a function of two parameters — at each corner, one headword is defined as a residual function of the other two: $A := \text{Res}(B, C)$. This is a function with two inputs.

Now suppose you have four concepts that seem mutually entangled. Naively, you might try to write something like $A := \text{Res}(B, C, D)$. This is a residual function of three parameters. The framework says this is too many. Laws are functions of two parameters, not three.

The resolution comes from the same move you would make in algebra or lambda calculus when you have a function with too many arguments: introduce an intermediate term. Factor the function. If $f(a, b, c, d)$ is your four-parameter relationship, look for a decomposition where $x = g(a, b)$ and the result $= h(x, c, d)$. The intermediate value x is a new concept — one that captures part of the relationship between the four terms and serves as a bridge between two smaller structures. In VD terms, x is the headword that was missing from your analysis. It becomes the point where two triplets overlap.

How to find the intermediate term. The intermediate term is the concept that captures what A and B “produce together” that C and D then “consume.” Ask yourself: of the four terms, which pairs seem to have a tighter relationship? What does the relationship between A and B create or establish that the relationship between C and D depends on? Name that thing. It might be a condition, a quantity, a structural property, a mode of behaviour. Whatever it is, it is your candidate for the intermediate headword.

Note that the intermediate term appears in both triplets — it is a winter word in the first and the November word of the second (or vice versa). This is exactly the overlapping house structure: the wall of the first triplet becomes the chimney of the second.

The Newton III example. Newton’s Third Law is the canonical case of this. It was traditionally presented as a single law relating action and reaction. The VD encoding revealed it to be two separate degree-4 laws sharing an intermediate concept. What looked like one four-cornered structure was two triplets overlapping at a shared term.

The general principle. Any time you have n terms that seem mutually entangled in a single law, look for the decomposition into overlapping triplets with $(n - 2)$ intermediate terms bridging them. The intermediate terms are the concepts your analysis has not yet named. Finding them is the hard part. But once found, the triplet decomposition will typically reveal structural features — degree differences, dependency asymmetries, hidden prerequisites — that were invisible in the monolithic n -term picture.